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Biography

Chenyu's research interests include simulation of mantle dynamics.

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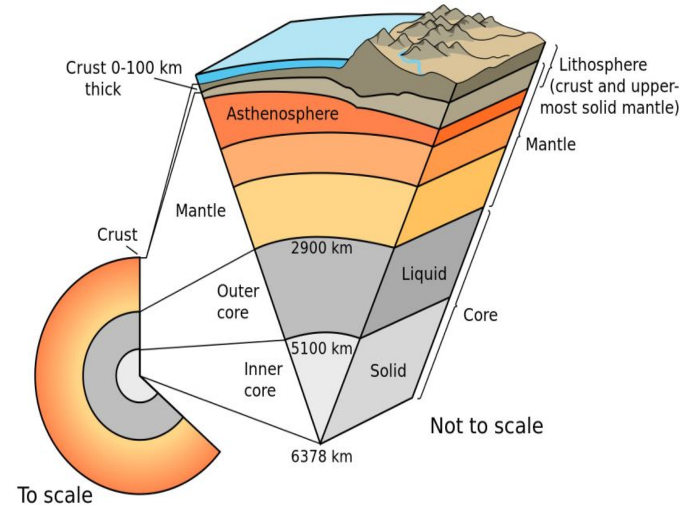
Partial melting in the asthenosphere is a crucial geophysical process in lithospheric evolution and plate tectonics.



(National Geographic)

Goal. Provide an accurate numerical approximation of the time-dependent dynamics of a partially molten mantle.

Objective.

- Provide a locally mass conservative numerical framework for
  1. The mechanics of a deformable solid matrix that is possibly partially molten.
  2. The transport phenomena of chemical species and energy in the presence of porous media.
- Connect static mechanical behavior and transport phenomena with proper phase behavior.

McKenzie Equations

(McKenzie 1984)

A two-phase formulation of a viscous mantle.

$$
\begin{aligned}
\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) &= 0, && \text{(Darcy)} \\
\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho}\mathbf{g} &= 0, && \text{(Stokes)} \\
\frac{\partial \rho_l \phi}{\partial t} + \nabla \cdot (\rho_l \phi \mathbf{v}_l) &= \Gamma \\
\frac{\partial \rho_s (1-\phi)}{\partial t} + \nabla \cdot (\rho_s (1-\phi)\mathbf{v}_s) &= -\Gamma
\end{aligned}
$$

Deviatoric stress of mixture

$$
\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s\phi_s\left(\tfrac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T) - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s \mathbf{I}\right)
$$

Unknowns.

Four equations and five unknowns.

- Velocities $\mathbf{v}_l$ and $\mathbf{v}_s$.
- Pressure p_l and p_s.
- Porosity ϕ / Melting source Γ.

Three parts.

- Static mechanic equations.
- Transport of chemical species and energy.
- Phase package or prescribed melting function.